In [ ]:
```
import pandas as pd
dataset=pd.read_csv("startup_dataset.csv")
dataset
```

In [ ]:
```
dataset=dataset.drop(columns=["Unnamed: 0"])
dataset
```

In [ ]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

In [ ]:

```
x=dataset[["RD Spend","Administration","Market Spend","State_Florida","State_New York"]]
y=dataset[["Profit"]]
print(x)
print(y)
```

In [ ]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
y_test
```

In [12]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [13]:
```
from sklearn.tree import DecisionTreeRegressor
r=DecisionTreeRegressor(
    criterion='squared_error',splitter='best', max_depth=None,
    max_features=1000, random_state=42, max_leaf_nodes=12,
    min_impurity_decrease=0.0,
)
r.fit(x_train, y_train)
```

Out[13]:

▾ DecisionTreeRegressor ⓘ ?

▸Parameters

▸Fitted attributes

In [15]:
```
y_pred=r.predict(x_test)
y_pred
```

Out[15]:

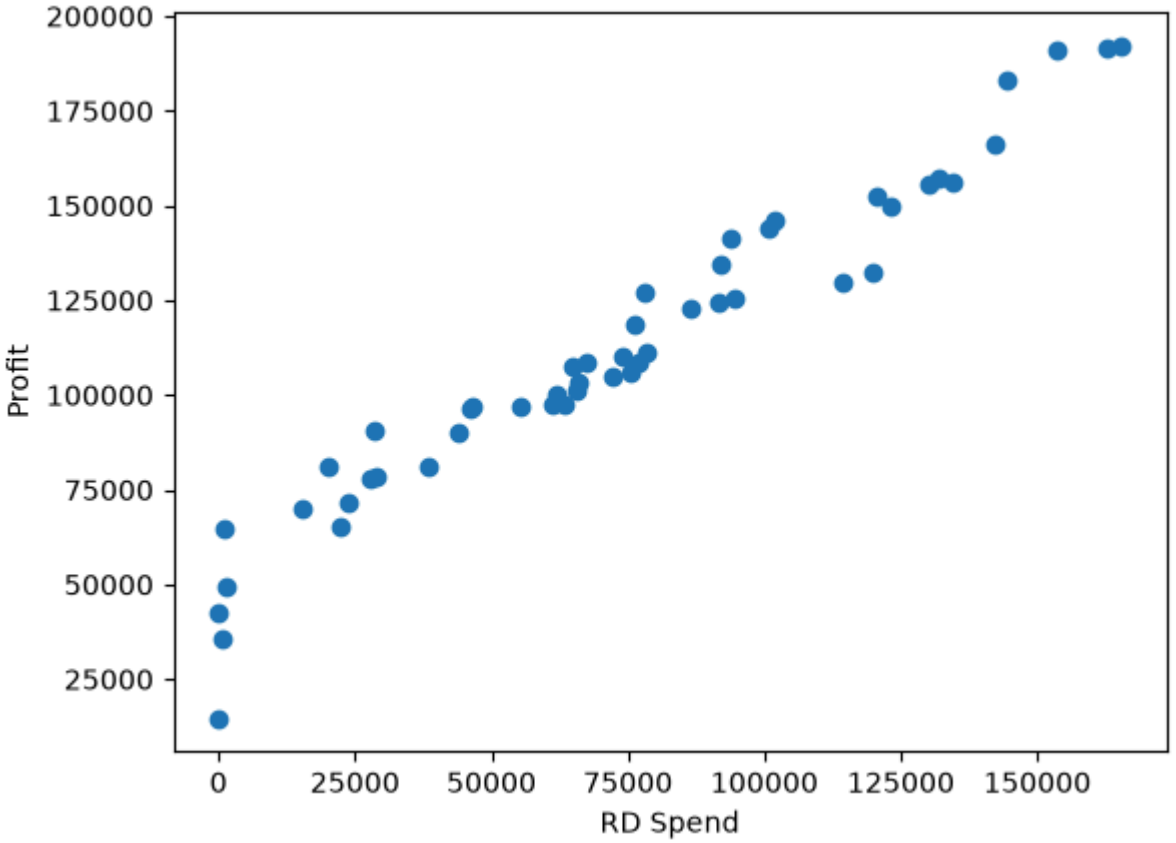
```
array([143989.05666667,  86299.18      ,  97691.986     ,  49490.85      ,
       128928.96333333,  42559.83      , 116333.105     ,  97691.986     ,
        97691.986     , 143989.05666667])
```

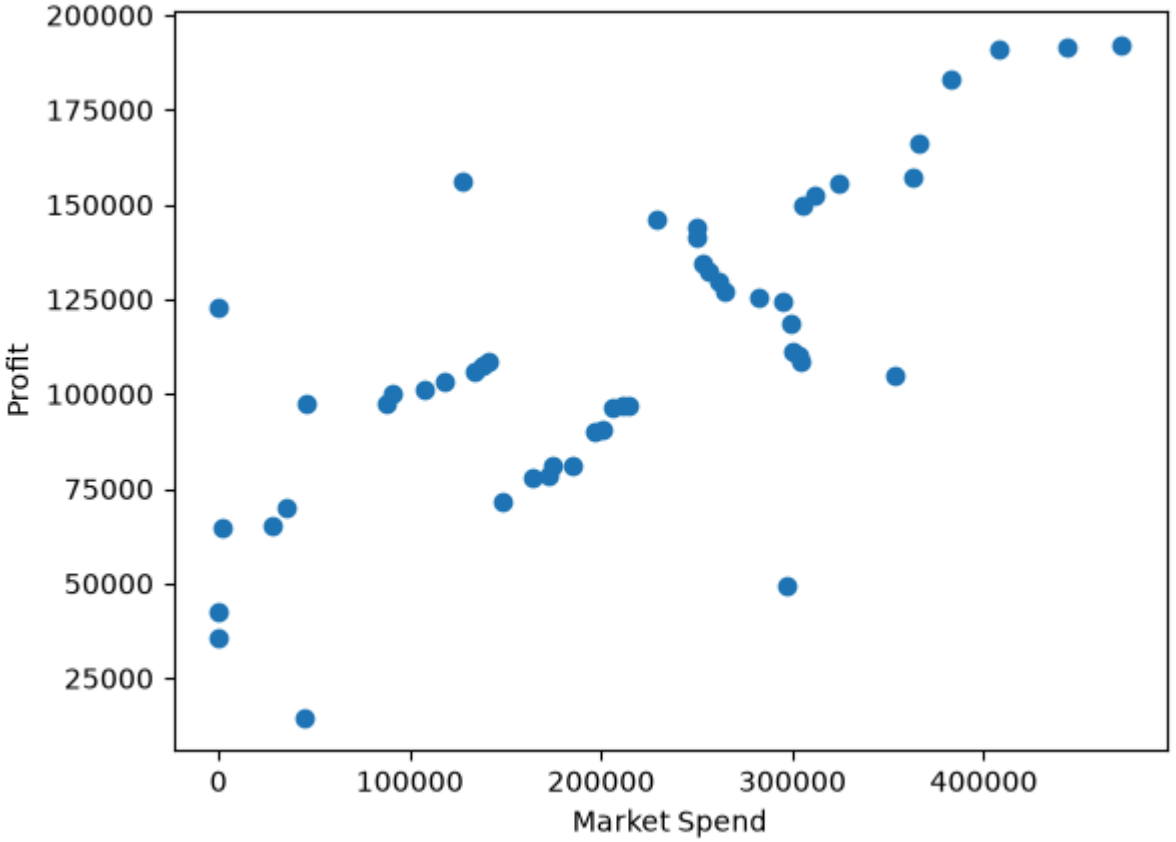
In [16]:
```
from sklearn.metrics import r2_score
result=r2_score(y_pred,y_test)
result
```

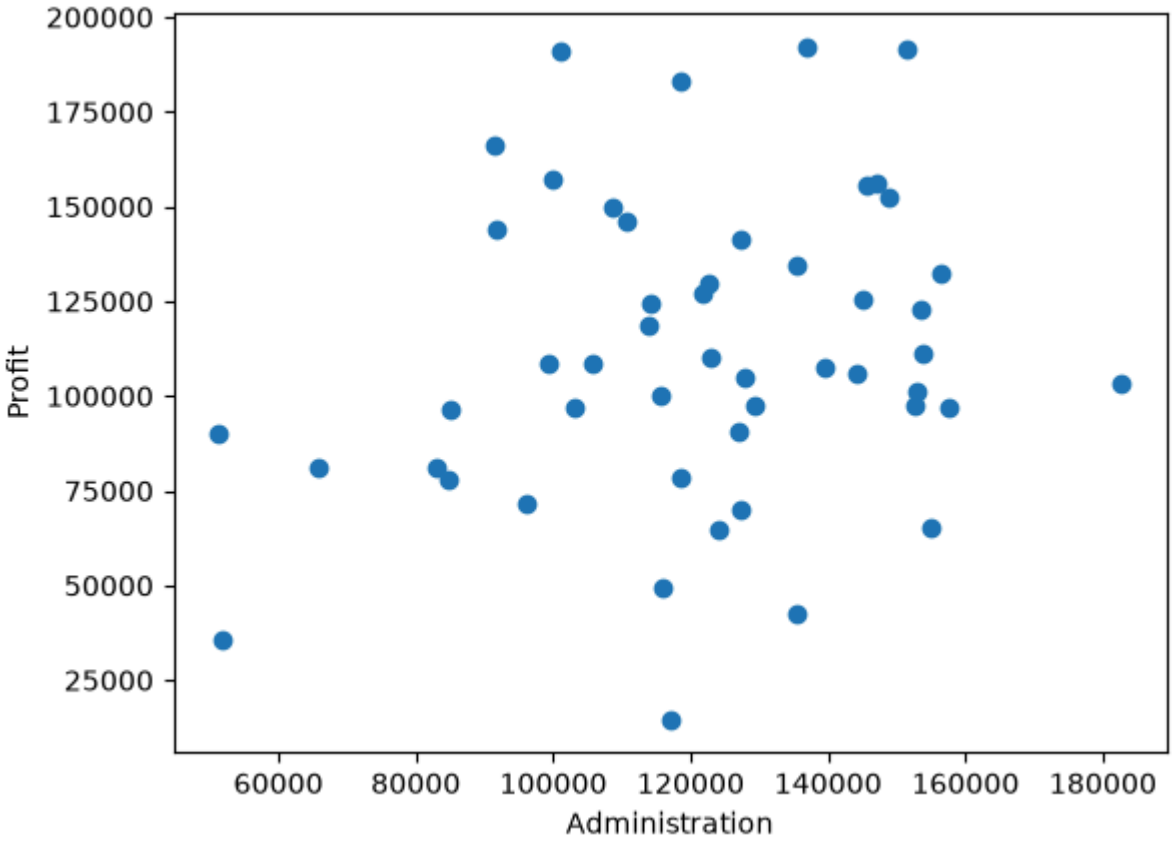
Out[16]:

0.9018816319515847

In [17]:
```
import matplotlib.pyplot as plt
plt.scatter(dataset["RD Spend"],y)
plt.xlabel("RD Spend")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Market Spend"],y)
plt.xlabel("Market Spend")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Administration"],y)
plt.xlabel("Administration")
plt.ylabel("Profit")
plt.show()
```




200000
175000
150000
125000
100000
75000
50000
25000
Profit
0
100000
200000
300000
400000
Market Spend


200000
175000
150000
125000
100000
75000
50000
25000
Profit
60000
80000
100000
120000
140000
160000
180000
Administration

In [ ]: